

Chapter 11: Cell Structure and Micro-organisms

Textbook Exercise Solutions & Answers



1. Answer the following questions.

(a) What is a cell?

The cell is the fundamental, structural and functional unit of living organism.

(b) Name the different organelles in a cell.

The nucleus, endoplasmic reticulum, Golgi bodies, lysosomes, mitochondria, vacuoles, plastids, chloroplasts are the different organelles in a cell.

(c) What are micro-organisms?

The organisms which cannot be seen with our eyes but can only be observed under a microscope are called micro-organisms.

(d) Which are the different types of micro-organisms?

Algae, fungi, bacteria, protozoa and viruses are various types of micro-organism. Some are unicellular and others are multicellular. Some of them are useful and some are harmful.

2. Fill in the blanks with the proper word.

(a) The organelle called the is present in the plant cells only.

chloroplasts

(b) Garbage is converted into by microorganisms.

fertilizer

(c) In the cell, photosynthesis is carried out with the help of

chloroplast

(d) An electron microscope is necessary for the study of

micro-organism

(e) The process of preparing their own food in presence of sunlight and chloroplast by plants cells is known as

photosynthesis.

3. What is the difference between us?

(a) Plant cell and animal cell.

Plant cell	Animal cell
1. Plant cell has cell wall.	1. Cell wall is absent in animal cell.
2. Plant cell has definite shape due to presence of cell wall.	2. Animal cell do not have definite shape.
3. Plant cell has one big vacuole.	3. Animal cells have many small vacuoles.
4. Plant cell has chloroplast so they produce their own food.	4. Animal cells do not have chloroplast, so they cannot produce their own food.
5. Plant cell has plastids.	5. Plastids are absent in animal cell.
6. Centrosomes are absent in plant cell.	6. Centrosomes are present near the nucleus.
7. Lysosomes are absent in plant cell.	7. Lysosomes are present in animal cell.

(b) Prokaryotic cell and Eukaryotic cell

Prokaryotic cell	Eukaryotic cell
1. No well defined nucleus.	1. Well defined nucleus with a nuclear membrane
2. Ribosomes are small.	2. Ribosomes are large.
3. A single length of DNA is present.	3. Several lengths of genetic materials (DNA) is present.
4. No other cell organelles are present.	4. Several organelles like mitochondria, chloroplasts, etc. are present.
5. Examples: Bacteria, blue green algae	5. Examples: Euglena, Amoeba, all plant and animal cells

4. Sketch and describe in your own words, the plant cell and animal cell.

Sketch and describe in your own words, the plant cell and animal cell.

There are two types of cell—animal cells and plant cells. These cells contain various types of membrane beyond cell-organelles. Plant cells have a definite shape due to the presence of cell wall. Besides, unlike animal cells, plant cells contain single large vacuole. All these cells are known as eukaryotic cells. It cell has four main parts: the cell wall, cell membrane, cytoplasm and cell organelles.

1. Cell wall: The cell wall is the outermost covering of a cell. It is present only in plant cells.
2. Plasma membrane: The plasma membrane, also called the cell membrane, is a kind of thin covering, (a) It is extremely delicate and flexible, (b) It is the outermost covering of animal cells.
3. Cytoplasm: The liquid part in the cell, present around the nucleus is called cytoplasm, (a) It occupies the space between the plasma membrane and nucleus, (b) Cell organelles are scattered in the cytoplasm.
4. Cell organelles: These mainly include the nucleus, endoplasmic reticulum, Golgi bodies, lysosomes, mitochondria, vacuoles, plastids etc.

Plant cells contain chloroplasts.

(a) The nucleus is the most important organelle of the cell, (b) There is a porous double membrane around it. (c) The nucleus controls all functions of the cell, (d) The endoplasmic reticulum is a sprawling net-like organelle. Its function is to make necessary changes in the proteins produced by ribosomes and send them to the Golgi bodies, (e) Golgi bodies are made up of several flat sacs. Their function is the proper distribution of proteins.

(f) Mitochondria and plastids are organelles with double outer coverings, (g) As mitochondria produce energy, they are called the powerhouses of the cell, (h) The chloroplasts in plant cells carry out the function of photosynthesis, (i) Vacuoles help to throw out waste products of the cell, (j) Vacuoles in animal cells are small whereas there is only one large vacuole in a plant cell.

5. Explain the uses and the harmful effects of micro-organisms.

Micro-organisms which are helpful are called useful micro-organisms.

Uses:

1. Micro-organisms present in nodules of leguminous plants, convert atmospheric nitrogen into its compounds and these compounds help to increase the soil fertility and protein content in the soil.
2. Some micro-organisms are useful in process of fermentation, making the food easily digestible and more nutritious. For producing milk products like butter, buttermilk, cheese, paneer, etc.
3. Microbes present in soil, dung etc. decompose garbage and convert into manure of best quality and surrounding is kept clean.
4. Microbes are also useful for sewage disposal. They help in decomposition of organic compounds in it.
5. Microbes are used in production of vaccines.
6. Microbes are also used in processes like tanning of skin, production of ropes and strings, from agave.
7. Some microbes use oil for their growth. Such microbes are used to clear a layer of oil floating on the surface of an ocean or lake formed due to leak or spill.
8. Farm waste, human urine and faeces, wet garbage etc. is collected and used in a biogas plant to produce biogas and fertilizer.

Harmful effects:

1. Disease producing micro-organisms are called pathogens, they are harmful micro-organisms.
2. Fungus grows on moist food, and microbes release enterotoxins into the food. Such toxins spoil the foods. Eating such spoiled food causes food poisoning i.e. loose motions and vomiting.
3. Pathogens may be present in water bodies contaminated with sewage, and dirt from surroundings, in food with flies sitting on it. If such contaminated food or water is consumed, we may fall ill with diseases like amoebiasis, typhoid, cholera, hepatitis, gastro, etc.
4. Pathogens are released in the air by a person having infection of the respiratory, sneezes or coughs. A healthy person may get infected by pathogens on breathing in the same air and contract diseases like common cold, cough, diphtheria, pneumonia, tuberculosis, etc.
5. Microbes that cause diseases like malaria, dengue, elephantiasis, yellow fever, chikunguniya, zika fever etc. gain entry into the human body through the bite of a female mosquito.

6. Give reasons.

(a) Diseases spread on a large scale during periods of heavy rainfall and floods.

1. Due to floods there is shortage of clean, safe drinking water.
2. It gets contaminated with sewage and dirt having disease causing pathogens, causing disease of alimentary canal, typhoid, cholera, dysentery.
3. Due to stagnant water, many mosquitoes breed on it and cause diseases like malaria, dengue, etc.

(b) There is a possibility of food poisoning if we eat stale food.

1. Fungus grows quickly on moist and stale food. Micro-organisms grown on the food release toxic materials like enterotoxin into the food.
2. Such toxins spoil the food.
3. Eating such spoiled food may cause loose motions and vomiting. So, we should eat food which is freshly prepared.

(c) Soil is turned over during tilling.

1. Tilling is done to loosen the soil in initial agricultural practice.
2. Loose soil allows the penetration of roots of the plant.
3. By turning of soil, it allows uniform mixing of manure and fertilizer.
4. It increases the fertility of soil. Seeds also germinate easily in aerated soil. Hence, soil is turned during tilling.

(d) Fungus grows quickly in moist or humid condition.

1. Fungus needs water and moisture for its growth.
2. Humidity brings about spore germination and the fungus is able to grow and reproduce quickly in moist or humid condition.

(e) A refrigerator is used in almost every home.

1. Refrigerator is used to keep food item in healthy condition and to keep them fresh.
2. If the food items are kept outside, they spoil very fast due to growth of micro-organisms and they make food spoiled and harmful.
3. The optimum temperature for the growth of micro-organism is 15°C to 35°C.
4. In refrigerator the temperature is kept very low. So, the food is preserved in refrigerator.

(f) Bread rises during baking.

1. The process of fermentation is used in baking, for example, in making bread. Yeast is added to make the dough rise.
2. The fermentation is a chemical process of conversion of one type of carbon compound into another type of carbon compound by the action of micro-organisms.
3. Heat is generated in this process. Carbon dioxide (CO₂) and some other gases are released.
4. These gases cause an increase in volume so the bread dough rises.

(g) Fodder is soaked in water before offering to cattle.

1. In dry fodder only some part of it is useful for cattle and rest is wasted.
2. When fodder is soaked in water, it sprouts.
3. So, to make the fodder more nutritious and more easy to digest it is soaked in water before offering to cattle.

7. When will you use a simple microscope and when a compound microscope?

1. I will use a simple microscope to magnify the objects and to see the growth of a fungi or mucor on a piece of moist bread.
2. I will use a compound microscope in Research Laboratory and high school science lab to observe micro-organisms which we are not able to see with our naked eye.